BASIC SQL QUERIES AND AGGREGATE FUNCTIONS

ASSIGNMENT 8

DATABASE MANAGEMENT SYSTEM

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Introduction

SQL queries are used to retrieve, filter, sort, and analyze data stored in a database. SQL clauses such as WHERE, ORDER BY, GROUP BY, and HAVING help organize and filter records. Aggregate functions such as COUNT(), SUM(), AVG(), MIN(), and MAX() perform calculations on data. In this assignment, these SQL commands are applied to the Food Delivery Management System using MySQL.

1. Filtering Records Using WHERE

The WHERE clause is used to retrieve records that satisfy a specified condition.

```
SELECT Item_ID, Item_Name, Price
FROM MENU_ITEM
WHERE Price > 150;
```

Output:

Item_ID	Item_Name	Price
301	Paneer Biryani	220.00
305	Chicken Tandoori	280.00
308	Veg Fried Rice	170.00

This query displays menu items whose prices are greater than ₹150. Only records that satisfy the condition are displayed.

2. Sorting Records Using ORDER BY

The ORDER BY clause sorts records in ascending (ASC) or descending (DESC) order.

```
SELECT Restaurant_Name, Rating
FROM RESTAURANT
ORDER BY Rating DESC;
```

Output:

Restaurant_Name	Rating
Urban Tandoor	4.7
Spice Garden	4.5
Green Bowl	4.2

South Kitchen	4.0
Food Junction	3.9

This query displays restaurants in descending order of their ratings. The restaurant with the highest rating appears first.

3. Grouping Records Using GROUP BY

The GROUP BY clause groups records that have the same value in a specified column. It is often used with aggregate functions.

```
SELECT Order_Status, COUNT(*) AS Total_Orders
FROM ORDERS
GROUP BY Order_Status;
```

Output:

Order_Status	Total_Orders
Delivered	3
Preparing	1
Confirmed	1

This query groups orders according to their status and counts the number of orders in each group.

4. Filtering Grouped Results Using HAVING

The HAVING clause filters groups created by GROUP BY, usually using an aggregate condition.

```
SELECT Restaurant_ID, COUNT(*) AS Total_Menu_Items
FROM MENU_ITEM
GROUP BY Restaurant_ID
HAVING COUNT(*) >= 2;
```

Output:

Restaurant_ID	Total_Menu_Items
101	2
102	2
104	2

This query groups menu items by restaurant and displays only restaurants that have at least two menu items. Restaurant IDs 101, 102, and 104 satisfy the condition.

5. Aggregate Functions

Aggregate functions perform calculations on a set of records and return a single result for each calculation.

A. COUNT()

The COUNT() function counts the number of records in a table or group.

```
SELECT COUNT(*) AS Total_Customers
FROM CUSTOMER;
```

Output:

Total_Customers
5

This query counts the total number of customer records in the CUSTOMER table.

B. SUM()

The SUM() function calculates the total of numeric values in a column.

```
SELECT SUM(Amount) AS Total_Payment_Amount
FROM PAYMENT;
```

Output:

Total_Payment_Amount
970.00

This query calculates the total amount across all five payment records, including payments marked as pending.

C. AVG()

The AVG() function calculates the average of numeric values in a column.

```
SELECT AVG(Rating) AS Average_Rating
```

FROM RESTAURANT;

Output:

Average_Rating
4.26

This query calculates the average rating of the five restaurants in the database.

D. MIN()

The MIN() function returns the smallest value in a column.

SELECT MIN(Price) AS Minimum_Price

FROM MENU_ITEM;

Output:

Minimum_Price
40.00

This query identifies the lowest menu item price, which is ₹40.

E. MAX()

The MAX() function returns the largest value in a column.

SELECT MAX(Price) AS Maximum_Price

FROM MENU_ITEM;

Output:

Maximum_Price
280.00

This query identifies the highest menu item price, which is ₹280.

Conclusion

In this assignment, basic SQL queries and aggregate functions were applied to the Food Delivery Management System database. The WHERE clause filtered records, ORDER BY sorted data, GROUP BY grouped records, and HAVING filtered grouped results. The aggregate functions COUNT(), SUM(), AVG(), MIN(), and MAX() were used to calculate useful information about customers, payments, restaurants, and menu items. These queries help retrieve meaningful information and support data analysis in the system.